

Internship Report

To Determine the Small G Protein, RhoG Expression in Response to ILK in Human Endometrial Epithelial Cells

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Abstract

Integrin-linked kinase is a protein crucial in cell signaling, particularly in cell-matrix interactions. It is involved in various intracellular signaling pathways that regulate cell growth, survival, adhesion, migration and invasion. RhoG is a small monomeric GTPase of the Rac subfamily of Rho GTPases is an important component of various intracellular signaling pathways regulating cell motility, adhesion and recruitment. ILK and RhoG are functionally related through their involvement in cell migration and polarization, they interact in a ternary with ELMO2 forming the ERI complex crucial for Rac1 activation and subsequent processes. We overexpressed ILK in human endometrial epithelial cells and checked its effect on RhoG which resulted in increased levels of RhoG in the cells suggesting their direct involvement in regulating each other's expression.

Visual Abstract

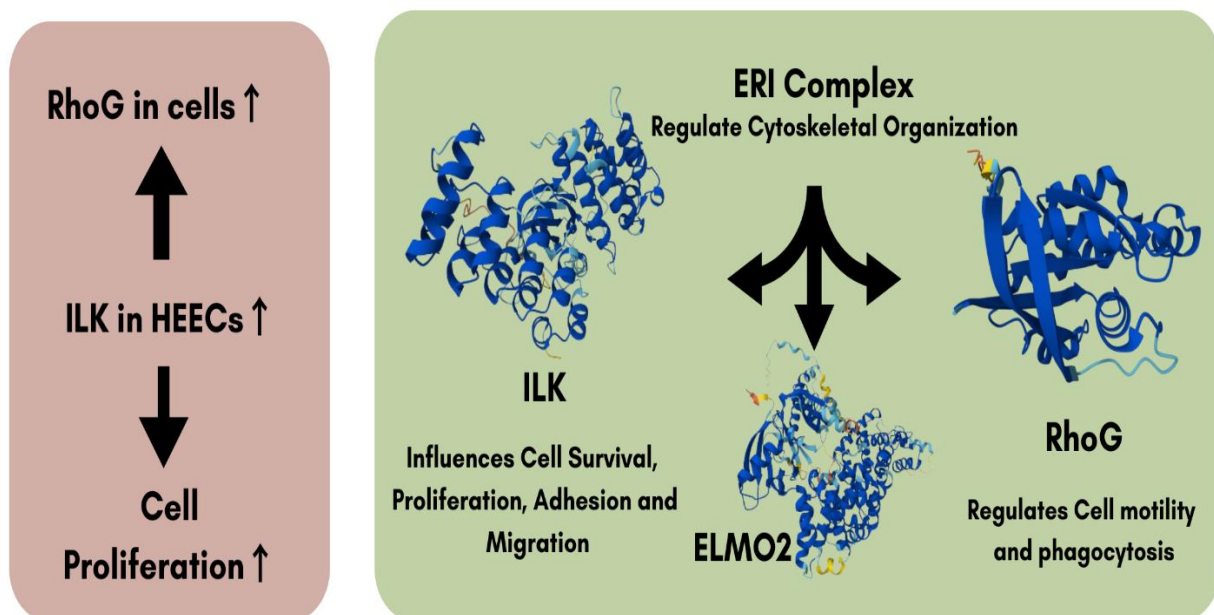


Figure 1

Structures: <https://alphafold.ebi.ac.uk/entry/Q13418>, <https://alphafold.ebi.ac.uk/entry/P84095>, <https://alphafold.ebi.ac.uk/entry/Q96JJ3>

Introduction

The Ishikawa cell line, first described by Nashida et al., is known for being an excellent model for the understanding of normal endometrial epithelium as it maintains estrogen and progesterone receptors as well as expresses many of the same enzymes and proteins found in normal endometrium (Athur J. Castelbaum, 1997). These well differentiated human endometrial epithelial cell lines maintain the same constitutive integrins which are noted in normal endometrial epithelium (Athur J. Castelbaum, 1997). Integrins are useful phenotypic markers of endometrial function and play crucial roles in normal implantation, uterine receptivity making them a great biological tool for gynecological studies (Athur J. Castelbaum, 1997).

Integrin-linked kinase (ILK), a serine-threonine kinase, is ubiquitously expressed and evolutionally conserved multifunctional adaptor protein which connects integrins to the actin cytoskeleton and recruits other adaptor molecules for the regulation of multiple cellular processes (Zhaopeng Yana, 2014). It is a crucial signaling pathway that participates in various processes mediating the relationship between extracellular matrix and cell-cell interaction and regulates cell proliferation, survival, migration and invasion (Qiao-Mei Zheng, 2018), (Qiaomei Zheng, 2016). Increased expression of ILK facilitates tumor metastasis in addition to promoting the pathogenesis and development of tumors (Qiao-Mei Zheng, 2018). Overexpression of ILK enhances migration and invasion abilities of endometrial stromal cells by facilitating the Epithelial-Mesenchymal Transition of cells. Epithelial-mesenchymal transition leads epithelial cells to lose cell polarity and are converted into mesenchymal cells leading to enhanced survival, extracellular matrix production, cytoskeleton modulation, invasiveness and malignancy (Athur J. Castelbaum, 1997). ILK overexpression is often correlated with various malignancies including non-small cell lung cancer, colorectal cancer, renal cell carcinoma, endometriosis etc. These conclusions are also supported by the results obtained by studying the effects of downregulation of ILK (Qiao-Mei Zheng, 2018). However, the functional role of ILK in carcinogenesis is tumor type-specific as it also functions as a tumor suppressor in breast cancer, aggressive pediatric tumor and rhabdomyosarcoma (Qiaomei Zheng, 2016).

Rho GTPases belong to the Ras superfamily of small GTPases, they act as molecular switches in the regulation of a wide array of cellular processes, including axon guidance, cell polarity, cell motility, vesicular trafficking and cell cycle progression (Shubham Kumar Rai, 2023). The Rho family consists of 20 members in humans classified into Cdc42 subfamily (RhoJ, RhoQ and Cdc42), Rac subfamily (RhoG, Rac1, Rac2 and Rac3), Rho UV subfamily (RhoU and RhoV), Rho A subfamily (RhoA, RhoB and RhoC), Rnd subfamily (Rnd1, Rnd2 and Rnd3), RhoD subfamily (RhoD and RhoF) and RhoBTB subfamily (RhoBTB1, RhoBTB2 and RhoH) (Shubham Kumar Rai, 2023) (Elena Vigorito, 2003). RhoG (Ras Homology Growth-related or Ras Homology family member-G) is a part of the Rac subfamily, is a small monomeric G protein (GTP-binding protein), shares ~70% sequence similarity with other members of the Rac subfamily and ~60% with Cdc42 subfamily of GTPases (Shubham Kumar Rai, 2023) (Elena Vigorito, 2003). Rac subfamily

members typically regulate membrane ruffling and lamellipodia formation upon interaction with effector proteins, whereas RhoG is known for regulating cellular signaling, gene transcription and other immunological diseases and abnormality contributing processes (Shubham Kumar Rai, 2023). Rho GTPases are crucial in regulating adhesion and migration but their expression in immune cells varies according to their developmental stage. RhoG is highly expressed in lymphocytes and activates cytoskeletal reorganization and gene transcription (Shubham Kumar Rai, 2023). It has a negative impact on immune responses by enhancing T-cell proliferation but plays an important role during T-cell activation on receptor-antigen contact in later phases (Shubham Kumar Rai, 2023). Role of RhoG in the regulation of neutrophil adhesion, migration and recruitment are also speculated but not enough work has been done in this niche.

Phagocytosis can be either an effective inflammatory defense performed or non-inflammatory phagocytosis of apoptotic cells. F-actin polymerization is common to all phagocytosis, leads to the formation of the phagocytotic cup, in which RhoA and its isoforms stabilize F-actin on activation while Rac1 and its isoform Rac2, Rac3 and RhoG modulates F-actin formation (Shubham Kumar Rai, 2023). Apoptosis can be induced by various perturbations that trigger respective program, small GTPases like Rho, Rac and Cdc42 are known stress-activated kinase stimulators leading to cell death however an interesting study by Murga et al. reported that Rac1 and RhoG protected fibroblast-like cell line from UV-induced apoptosis effectively (Shubham Kumar Rai, 2023). These findings demonstrate the ability of RhoG and Rac1 in activating both pro- and anti-apoptotic pathways.

Members of Rho family are key actin cytoskeleton regulators in diverse cellular functions. GDP-GTP exchange is required for Rho-family GTPase activation, which is catalyzed by Guanine nucleotide exchange factors, among them, Rac is activated at the leading edge of motile cells and induces actin-rich lamellipodia protrusion formation which are the major driving force for cell movement (Hironori Katoh, 2005). A study by Katoh et al. suggests that the activation of Rac1 and lamellipodia formation at the leading edge is attenuated in migrating RhoG-knockdown cells suggesting that RhoG is a key upstream regulator for Rac during cell migration and induces extensive membrane ruffling at the cell periphery in response to fibronectin (Hironori Katoh, 2005).

Studies have shown that forced expression of ILK enhanced migration abilities of cancer cells, while silencing of ILK inhibited tumor growth and invasion suggesting a direct role of ILK in cell proliferation and migration in tumors (Qiaomei Zheng, 2016). Upregulation of ILK and increased migration and invasion abilities were observed in endometrial stromal cells of endometriosis suggesting that targeting ILK expression can aid in the discovery of therapeutics for tumors, endometriosis and various other diseases (Qiaomei Zheng, 2016). Overexpression of ILK in colorectal cancer cells showed contributions in EMT-like transformation, invasion and migration in CRC cells (Zhaopeng Yana, 2014).

Recent studies suggest that microtubules may serve as targets for Rho GTPases and coordinate global cytoskeletal responses. Rac1 activation in the context of cell migration results from the coordinated actions of several signaling pathways associated with complexes that contain ELMO2, RhoG and ILK (Bradley C. Jackson, 2015). RhoG is crucial in the regulation of cell migration through ELMO- and Dock 180-dependent activation of Rac1, however, other signaling pathways in cell migration promotion by RhoG might also be involved as other RhoG mediated morphological events involve Cdc42 and Rac1 activation (Hironori Katoh, 2005). In the migration of undifferentiated keratinocytes, active RhoG recruits cytoplasmic species containing ELMO2 and ILK to cell front which results in the formation of ternary complex where ELMO2 bridges active RhoG and ILK, is termed ERI complex (Bradley C. Jackson, 2015). ERI complexes are involved in Rac1- and $\beta 1$ integrin dependent development of front-rear polarity and cell migration in response to epidermal growth factor, disruption of these complexes by depleting ILK or ELMO2 impairs these processes (Bradley C. Jackson, 2015).

In this study, we aimed to check and understand how ILK affects the expression of RhoG by overexpressing ILK in human endometrial epithelial cells as both play a direct role in mediating cell proliferation and migration. This analysis will be used in the understanding of their functioning and pathways to discover target-based therapeutics which will help diagnose several diseases including cancer and endometriosis.

Methodology

In-vitro

Cell Revival

A cell vial of Ishikawa cells (HG10736ANR) of cells in passage 36 preserved at -80 °C in DMSO were immediately put in boiled water after being taken out of cryopreservation. ~1mL of nutrient media MEM ALPHA X (Gibco, 12571-048) was added to the vial and centrifuged at 3000rpm for 3 minutes at 4 °C. The supernatant was discarded, and the pellet was dissolved in nutrient media rapidly.

Cell Culturing

T25 flasks were used for culturing the cells. The rich nutrient media of MEM ALPHA X composed of base medium of MEM supplemented with FBS (ThermoFisher Scientific, 12634010), NEAA (Gibco, 11140-050), Anti-Anti (ThermoFisher Scientific, 154240-016), non-essential amino acids (Gibco, 11360-070) were used for culturing the cells. The revived Ishikawa cells from cryopreservation were equally divided into 3 T25 flasks containing 3mL of MEM ALPHA X media and incubated at 36 °C with 5% CO₂ and humidified environment.

Subculturing

After 24 hours when the flask got confluent with cells, the media from the flasks was discarded and washed with PBS. 1mL trypsin was added to all the flasks and collected in an Eppendorf which was subjected to centrifugation at 3000rpm for 3 minutes. The supernatant was discarded, and the pellet was dissolved in nutrient media, which was divided into 2 new flasks. Hence, on subculturing, a total of 6 replicate flasks of Ishikawa cells were being cultured at 36 °C with 5% CO₂ in humidified conditions.

Transfection

The nutrient rich media was replaced with Opti-MEM (Gibco, 31985-050) to induce starvation in cells for 6 hours when the flask was ~90% confluent. In accordance with the manufacturer's instructions, a Lipofectamine 3000 reagent solution composing 9 µL Lipofectamine, 47 µL reagent and 94 µL Opti-MEM was prepared. Similarly, another DNA plasmid solution of 9 µL DNA sample and 141 µL Opti-MEM was prepared and incubated for 5 minutes. The transfection complexes were formed by mixing the two solutions and further incubation of 30 minutes. The cells were transfected by incubating them in this complex with Opti-MEM media. Morphological changes were visualized under a microscope and images were captured by a microscope equipped with a digital camera (DSFi2, and DSU3, Nikon, Japan).

Protein Isolation

Hypoxia treatment was given to the cells for 4 hours. On washing the flask with PBS to remove the growth media, 4mL chilled PBS was added and the cells were scrapped using a scraper, the cells were collected. 500 μ L PBS was again added to the flask and cells were scraped and collected. The cells in PBS were centrifuged at 3000rpm for 3minutes at 4°C. The supernatant was discarded and 30 μ L RIPA buffer was added and cells incubated.

Protein Quantification

BCA Assay kit was used according to the instructions of the manufacturer for quantification of the total protein isolated from the ILK transfected Ishikawa cells. BCA reagent A was added with BCA reagent B to the 96-well plates. Protein was later added, and the concentration of the total protein content was calculated on a Nanodrop machine.

Western Blotting

SDS-PAGE

Optimized protocol was used for the whole preparation and procedure of Western Blotting. Initially a test gel was run to check for the efficiency of the reagents and Coomassie staining was done as the preliminary test for verification. 12% SDS-PAGE Resolving gel was prepared by the adding 2.35mL Double distilled water, 1.75 mL Tris HCl (pH 8.8), 2.80 mL 30% Acrylamide, 80 μ L 10% SDS, 80 μ L 10% APS and 6 μ L TEMED whereas, the stacking gel was composed of 1.525 mL Double distilled water, 625 μ L Tris HCl (pH 6.8), 325 μ L 30% Acrylamide, 70 μ L 10% SDS, 70 μ L 10% APS and 6 μ L TEMED. Stacking gel was cast over the resolving gel when the latter solidified and a 10 well comb was used to make wells. Protein sample was prepared by preparation of 10 μ L Laemmli Buffer, MQ and 20 μ g Protein sample, the mixture was boiled in water to denature the proteins and then pipetted into the wells along with one well containing a protein marker. The Gel was run in Running Buffer with composition; 3.0gm Trizma base, 14.4gm Glycine, 1.0gm SDS for 1L solution.

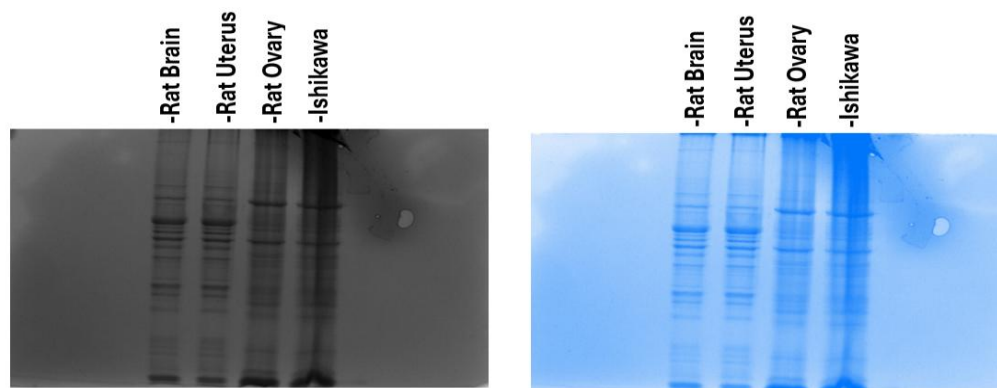


Figure 2: SDS-PAGE 12%; Lane Sequence: SD Rat Brain, SD Rat Uterus, SD Rat Ovary and Ishikawa; Protein Loading Concentration 20 μ g; Loading Volume: 10 μ L/Well; Running Voltage: 100V for 1.5 Hours; Staining with 0.1% Coomassie Brilliant Blue for 15 Minutes

Immunoblotting

In the apparatus of Electroblothing, a transfer sandwich was made with a sponge, sheets of blotting paper, PVDF membrane, SDS-PAGE Gel, Blotting sheets and again sponge. This setup was arranged into the transfer tank with the gel on cathode and the membrane on the anode side. Transfer buffer to facilitate the electroblothing process was prepared by adding 3.0gm Trizma base, 14.0gm Glycine and 200mL Methanol in 800mL Water for a 1L solution. For efficient transfer, a low temperature is required for which the setup was arranged in 4°C surroundings.

After transfer, the membrane was subjected to washing in TBST for 5 minutes three times. The blocking was done with 5% goat serum in TBST for an hour and the membrane was incubated in primary antibody (1:1000 concentration) overnight. The next day, after washing in TBST, the membrane was incubated in secondary antibody for an hour and was once again washed with TBST thrice.

Electrochemiluminescence

The membranes were visualized in Bio-Rad's ChemiDoc Imaging Systems Machine by Chemiluminescence. Ample amount of the chemiluminescence solution was poured on the membrane such that the membrane and hence the bands were soaked in the solution uniformly. The machine captured the images of the membrane with the band and were analyzed accordingly. Bands of β -actin (~42kDa) were used to validate that equal volume of protein sample were pipetted in the wells. The molecular weight of RhoG is ~21kDa, the bands must be observed in the respective region.

Statistical Analysis

The analyses were performed using three independent biological replicates. Nanodrop machine was used to quantify protein concentration from the proteins isolated from the human endometrial epithelial cells after performing BCA Assay. Cell size measurements were taken using ImageJ software for the qualitative analysis of morphology changes in cells. The reference scale was set with 124 pixels being equivalent to 500 μ m, this standard was estimated by measuring the length of the reference scale present in the clicked pictures. β -actin was used as loading control for the immunoblotting analysis. The significant differences between the two groups were calculated by unpaired T-test. The analyses were carried out using the GraphPad Prism version 5 (Boston, MA, USA).

Results

Morphological and RhoG Expression Analysis of HEECs on ILK overexpression

Both the control group and ILK overexpression group were transfected when the cells had covered ~90% space of the flasks. However, the flasks appear comparatively denser after transfection, even slightly more cells are observed floating in the medium. The cells in the transfected group appear to contain more nuclei and showed signs of increased mitotic activity. No major morphological differences were observed in either of the groups both before and after inducing overexpression.

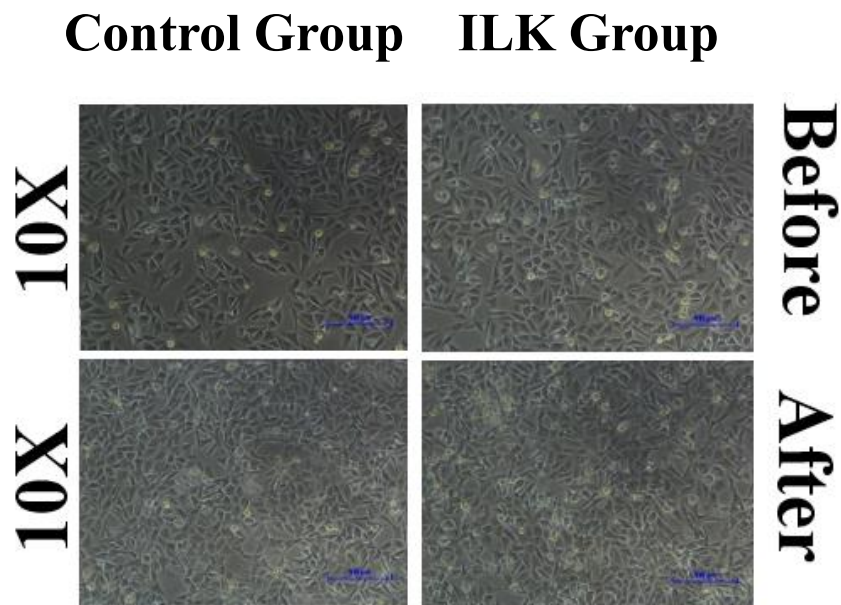


Figure 3: (A) Depicts the images of HEECs transfected with Control Plasmid group before transfection and after 48 hours of transfection. The cells exhibit increased proliferation post-transfection; (B) Compares ILK transfected group before transfection and after 48 hours of transfection. No significant morphological changes were observed except increased proliferation post transfection. ILK transfected group appears denser than the Control group

Our Western blotting analysis Revealed a band between 20-25kDa consistent with the molecular weight of RhoG (~21kDa), Bands between 37-50kDa corresponding to β -actin (~42kDa molecular weight) appeared uniform in size and thickness, indicating uniform protein loading across the wells. Notably, the protein bands of RhoG were darker in the ILK-overexpressing HEECs compared to the control vector HEECs, indicating upregulation of RhoG upon ILK-over-expression.

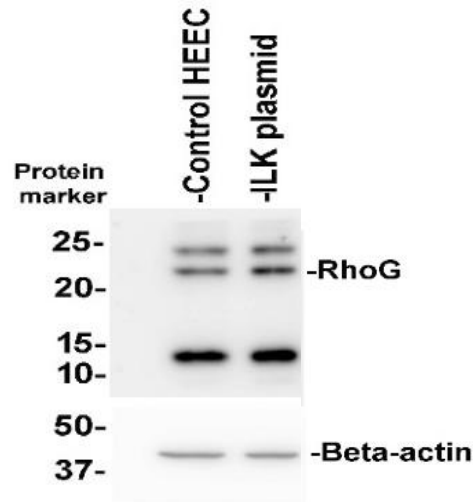


Figure 4: Illustrates western blot results following 12% SDS-PAGE gel. RhoG protein was detected using a primary antibody ab (dilution 1:1000) in 5% goat serum in TBST, incubated overnight. β -actin was used as a loading control. Bands corresponding to RhoG appear darker in ILK-transfected group

This observation was supported by an unpaired t-test analysis, which showed a statistically significant difference in RhoG expression between the two groups. Together these results demonstrate that ILK overexpression in human endometrial epithelial cells leads to increased RhoG expression.

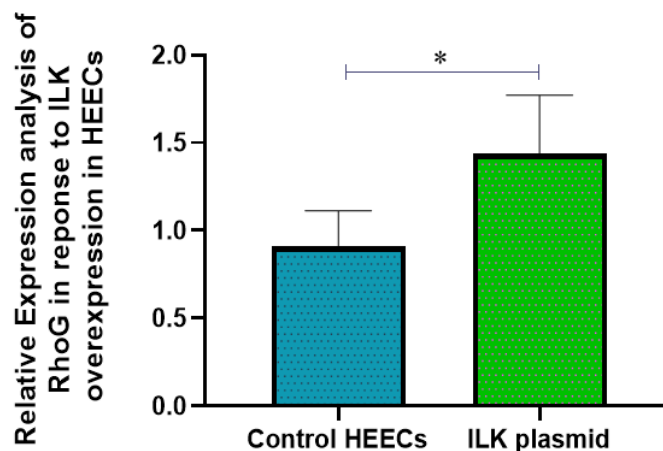


Figure 5: Statistical analysis revealed a significant upregulation of RhoG expression in the ILK-transfected cells on an unpaired t-test using GraphPad Prism.

Discussion

Cytoskeletal organization, proliferation and migration are crucial cellular processes required for normal functioning of cells. ILK is a multifunctional protein that acts as a kinase or a scaffolding protein, playing essential roles in cell differentiation, survival and adhesion, as it mediates intracellular signaling (Qiao-Mei Zheng, 2018). Whereas RhoG, a small GTPase of the Rac subfamily of Rho GTPases, is involved in cell migration, phagocytosis and endocytosis (Shubham Kumar Rai, 2023). Both ILK and RhoG are directly involved in ERI complexes bridged by ELMO8. ERI complex coordinates cell motility and cytoskeletal organization control (Bradley C. Jackson, 2015). Their functional interactions suggest a potential regulatory relationship between ILK and RhoG in cellular pathways.

In this study we investigated the effect of ILK overexpression on RhoG expression in human endometrial epithelial cells. Morphological analysis was first performed to assess any observable differences between the control group and ILK transfected group before and after transfection. Both groups showed increased cell proliferation post transfection, with ILK overexpressing group exhibiting slightly higher proliferation compared to the control group after transfection. No significant changes in cell size were observed, but the ILK group showed a modest increase in the number of floating cells probably due to insufficient space for adhesion and a greater number of cells undergoing division.

To quantify the effect of ILK overexpression on RhoG expression, western blotting was performed using protein lysate of the two groups. The bands of RhoG were visibly darker than the Control group. Densitometric analysis of the bands further confirmed this increase, and statistical analysis using an unpaired t-test revealed a significant difference between the two groups ($p < 0.05$).

Overall, the results suggest that overexpression of ILK upregulates the expression levels of RhoG in human endometrial epithelial cells indicating a direct or indirect relationship between the two to regulate the expression of each other. Further studies are required to elucidate the regulatory mechanisms involving ILK and RhoG, which may provide valuable insights for the development of targeted therapeutics in diseases such as cancer and endometriosis.

Conclusion

Our investigation indicates that ILK overexpression in the human endometrial epithelial cells leads to increased expression of RhoG. This supports the hypothesis that ILK and RhoG may functionally regulate each other within shared signaling networks.

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